Tables

1 $\frac{5}{36}$

2

a Sample Space $= \{0, 1, 2, 3, 4, 5\}$

b $\frac{1}{6}$

3

a

Spinner	2	3	4
7	9	10	11
9	11	12	13
12	14	15	16

b i $\frac{2}{9}$ ii $\frac{2}{9}$ iii $\frac{3}{9}$

4

a

	1	2	3	4	5	6
1	1,1	1,2	1,3	1,4	1,5	1,6
2	2,1	2,2	2,3	2,4	2,5	2,6
3	3,1	3,2	3,3	3,4	3,5	3,6
4	4,1	4,2,	4,3	4,4	4,5	4,6
5	5,1	5,2	5,3	5,4	5,5	5,6
6	6,1	6,2	6,3	6,4	6,5	6,6

b i $\frac{1}{18}$ ii $\frac{1}{36}$ iii $\frac{1}{6}$ iv $\frac{1}{2}$

5

a

	W	*X*	*Y*	*Z*
1	1, W	1, X	1, Y	1, Z
2	2, W	2, X	2, Y	2, Z
3	3, W	3, X	3, Y	3, Z
4	4, W	4, X	4, Y	4, Z
5	5, W	5, X	5, Y	5, Z
6	6, W	6, X	6, Y	6, Z

b 24

c i $\frac{1}{8}$ ii $\frac{1}{6}$ iii $\frac{11}{12}$

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6

a

	8	2	3	5
1	8	2	3	5
2	16	4	6	10
3	24	6	9	15
4	32	8	12	20
5	40	10	15	25
6	48	12	18	30

b $\frac{1}{4}$

c $\frac{1}{12}$

d $\frac{2}{3}$

7

a

	Cyan (C)	Pink (P)	White (W)
Black (B)	C, B	P, B	W, B
Grey (G)	C, G	P, G	W, G
Red (R)	C, R	P, R	W, R
Yellow (Y)	C, Y	P, Y	W, Y

b 12

c $\frac{1}{4}$

d $\frac{1}{3}$

8

a

	Labor (next election)	Liberal (next election)	Greens (next election)	Total
Labor (last election)	25	41	4	70
Liberal (last election)	16	73	1	90
Greens (last election)	9	26	5	40
Total	50	140	10	200

b $\frac{8}{45}$

c $\frac{1}{2}$

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d $\frac{57}{80}$

9

a

Spinner	2	4	5
3	5	7	8
7	9	11	12

b i 1 ii $\frac{5}{6}$ iii $\frac{2}{5}$ iv 1

Tree diagrams

10 $\frac{1}{9}$

11

a SR, SL, SC, GV, GM, GC

b $\frac{1}{6}$

c $\frac{1}{3}$

12

a $\{2, 3, 5, 7, 11, 13, 17, 19\}$

b i $\frac{1}{8}$ ii $\frac{1}{4}$

13

a $23, 24, 32, 34, 42, 43$

b i $\frac{2}{3}$ ii $\frac{1}{3}$ iii $\frac{1}{3}$

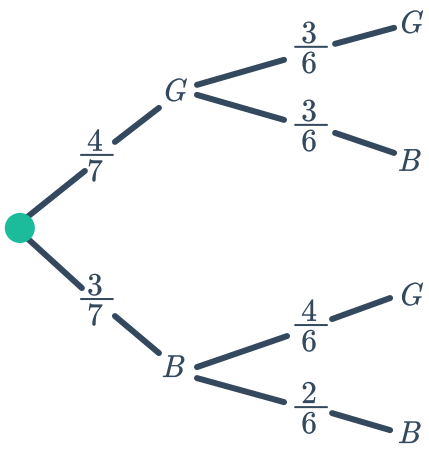
14

a 0.42 **b** 0.88

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15

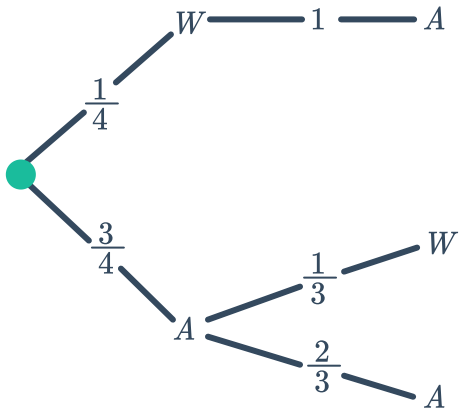
a



b i $\frac{3}{7}$ ii $\frac{2}{7}$

16

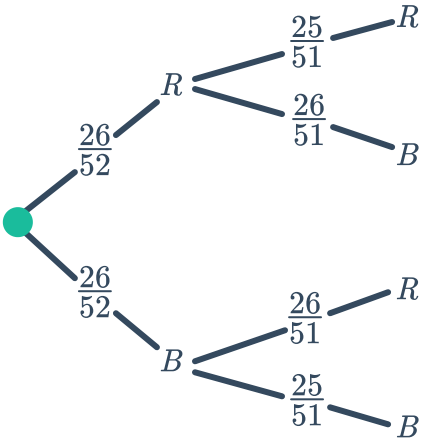
a



b i $\frac{3}{4}$ ii 1

17

a



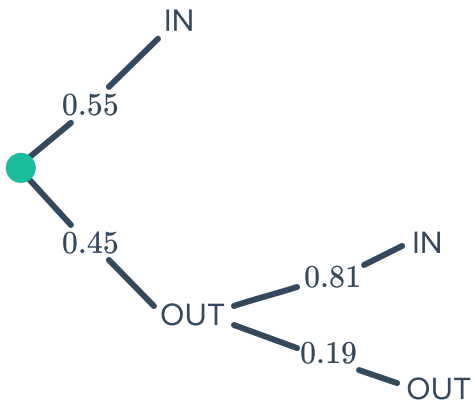
b $\frac{13}{51}$

c i Yes ii No iii No iv No v No

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18

a

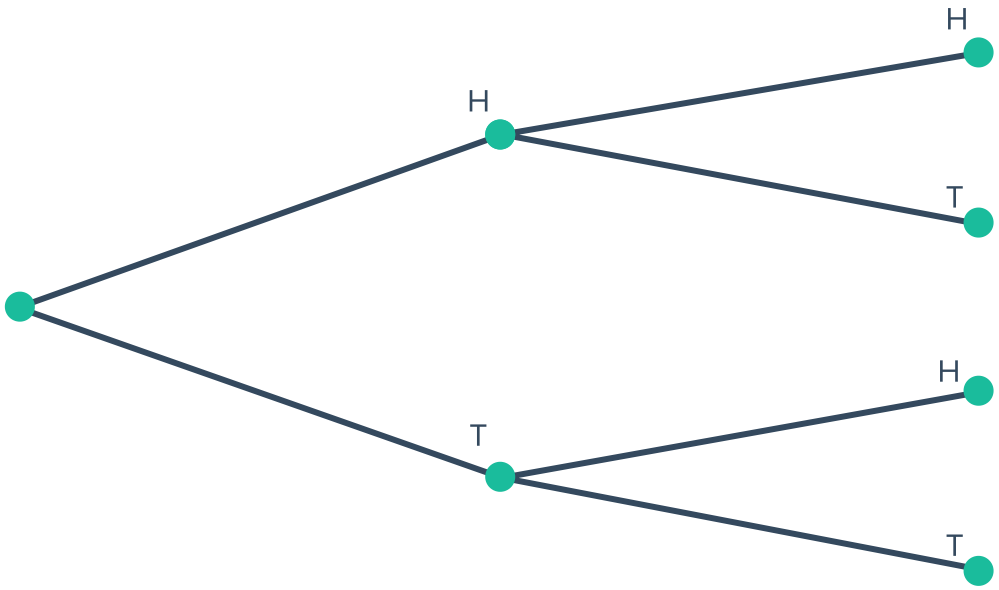


b

i 0.45 **ii** 0.0855

19

a



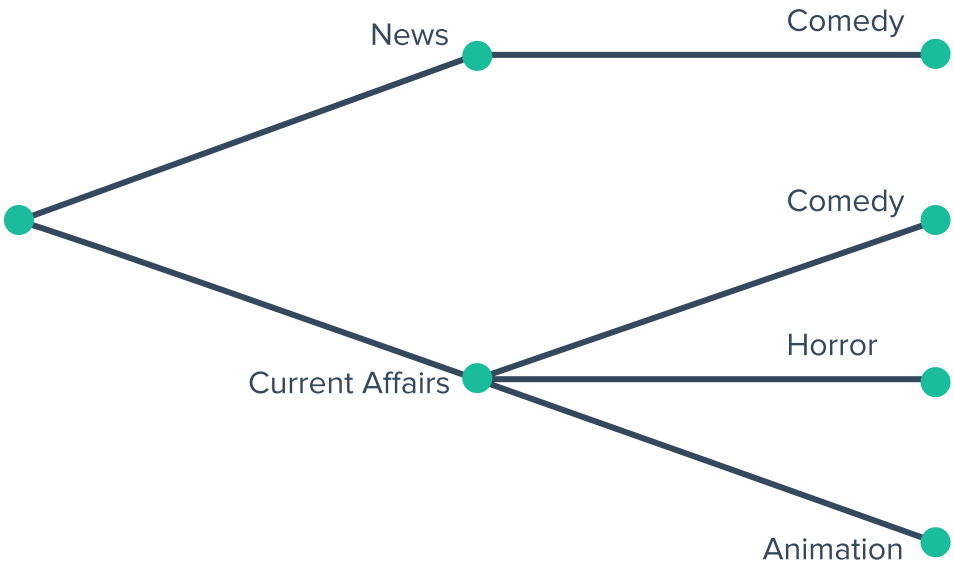
b

i $\frac{2}{4}$ **ii** $\frac{1}{4}$ **iii** $\frac{1}{4}$ **iv** $\frac{2}{4}$

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20

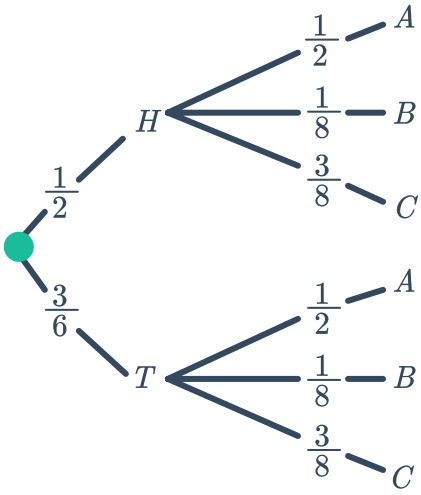
a



b $\frac{2}{3}$

21

a



b **i** $\frac{3}{16}$ **ii** $\frac{11}{16}$

22 $\frac{1}{7776}$

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23

a i $\frac{1}{9}$ ii $\frac{1}{27}$ iii $\frac{19}{27}$

b $\mathrm{P}(\text{Winning at least one of two games}) = \frac{5}{9}, \mathrm{P}(\text{Winning at least one of three games}) = \frac{19}{27}$ therefore Neville has a better chance at winning at least one of three games.

24

a i $\frac{1}{2}$ ii $\frac{1}{4}$ iii $\frac{1}{8}$ **b** $\frac{1}{2}$

c $\frac{1}{16}$

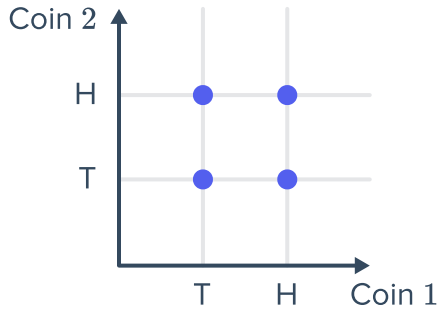
25

a $\frac{1}{2}$ **b** $\frac{1}{32}$

2D Grids

26

a

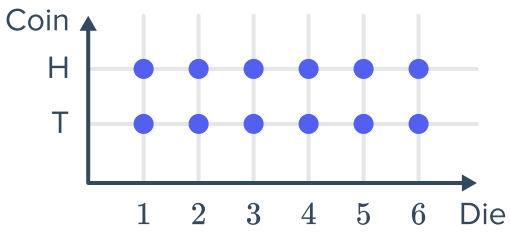


b 4

c $\frac{1}{2}$

27

a



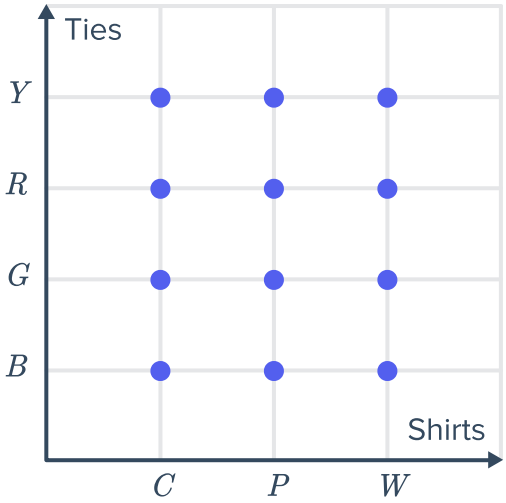
b 12

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28 **a**

	C	P	W
B	C,B	P,B	W,B
G	C,G	P,G	G,W
R	C,R	P,R	W,R
Y	C,Y	P,Y	W,Y

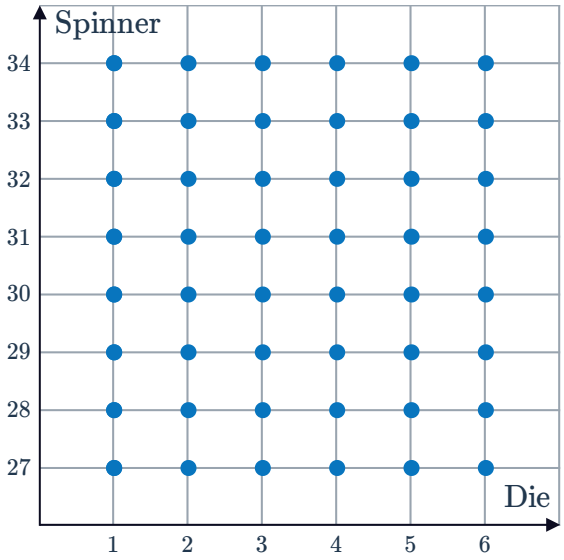
b



c 12

29

a



b 48

c $\frac{1}{4}$

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